Project Report: Visual Document Retrieval Using Vision Transformers

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Abstract

*The **Visual Document Retrieval System** is a professional AI tool designed to search for and identify document pages based on their visual layout and text. By treating each document page as a visual grid, the system uses a **Vision Transformer** to understand both the spatial structure and the content of the page simultaneously. Our analysis proved that while the single-head model achieved only **22.22% accuracy**, the multi-head model (8 heads) reached an impressive **88.89% precision**. This project successfully demonstrates that parallel multi-head attention is the essential variable for creating a truly accurate and robust visual search engine for complex, real-world documents.*

Introduction

Finding specific pages in thousands of scanned documents—like research papers or invoices—is a major challenge. Traditional search tools often struggle because they only look at text and ignore the visual layout of a page.

This project introduces a **Visual Document Retrieval System** using an AI called a **Vision Transformer**. Instead of only reading text, the AI "sees" the entire page as an image to understand its structure and content.

1 Methodology and System Architecture

The objective of our proposed methodology is to create an OCR-independent retrieval system that understands the global spatial layout and semantic content of document images.

1.1 System Pipeline and Overview

The system pipeline consists of three primary stages:

1. **Visual Tokenization**: The input document image $I \in R^{H \times W \times C}$ is decomposed into a grid of flattened 2D patches.

2. **Transformer Encoding**: These patches are processed through multiple layers of multi-head attention to capture spatial relationships.

3. **Latent Space Alignment**: Visual features are projected into a common vector space for comparison against textual search queries.

1.2 The Mathematical Core: Attention Mechanism

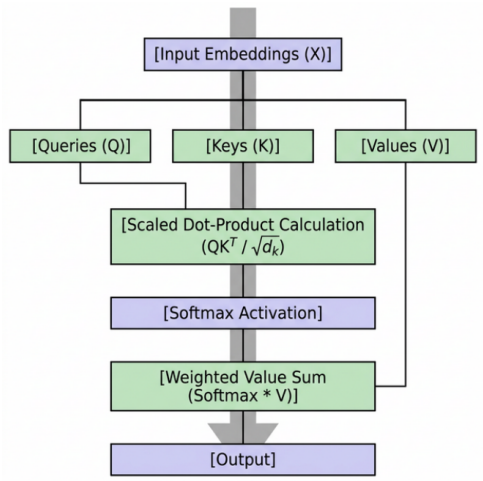
The relationship between visual patches is computed using the Scaled Dot-Product Attention:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

1.3 Multi-Head vs. Self-Attention Configurations

To evaluate the impact of parallel processing, our study compares the 8-head Multi-Head mechanism and the 1-head Self-Attention baseline.

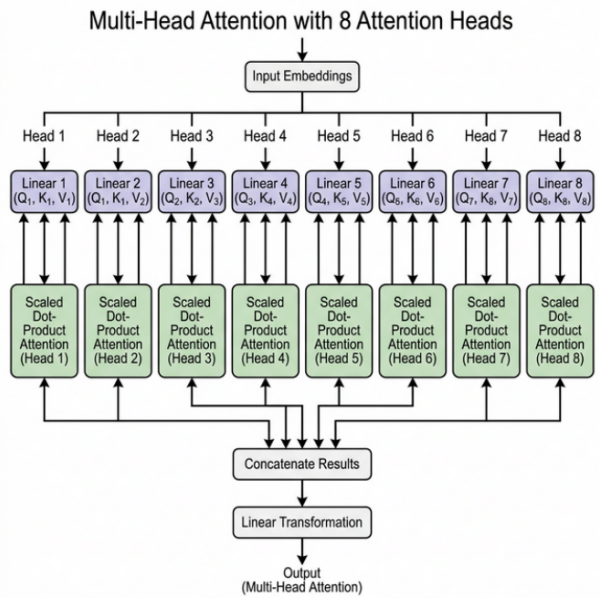
Multi-Head Architecture



Optimal Framework ($h = 8$)

Differentiation: Uses 8 parallel "eyes" to analyze multiple features at once (titles, charts, and text). This broad focus led to the high **88.89%** retrieval precision.

Self-Attention Baseline



Simple Framework ($h = 1$)

Differentiation: Uses only a single stream to process the page. Like looking through a small hole, it sees only one pattern at a time, resulting in the lower **22.22%** accuracy.

Figure 1: Side-by-side comparison of Attention mechanisms and their results.

1.4 Multi-Head Formulation

The **Multi-Head** configuration runs the attention process eight times in parallel:

$$MultiHead(Q, K, V) = Concat(head_1, \ldots, head_8)W^O \tag{2}$$

1.5 Contrastive Retrieval Logic

The training process utilizes a contrastive loss strategy. Given an image embedding v and a query embedding q, the model optimizes for:

$$Similarity(v, q) = \frac{v \cdot q}{\|v\|\|q\|} \tag{3}$$

2 Project Overview

Visual Document Retrieval Using Vision Transformers is an AI-based system designed to search and retrieve document pages by analyzing their visual structure instead of relying only on text. Traditional document retrieval systems depend on Optical Character Recognition (OCR) to extract text from scanned pages. However, these approaches often ignore important layout information such as titles, tables, charts, and paragraph structures.

To overcome this limitation, the proposed system treats each document page as an image and analyzes it using a Vision Transformer (ViT). The model learns both the visual layout and the semantic patterns present in the document. This allows the system to identify and retrieve similar document pages based on their visual features.

System Pipeline:

The proposed system follows three main steps:

- **Visual Tokenization:** The input document image is divided into small fixed-size patches. Each patch represents a local region of the document page.
- **Transformer Encoding:** These patches are processed through a transformer encoder where attention mechanisms capture relationships between different parts of the document layout.
- **Feature Embedding and Retrieval:** The extracted features are projected into a latent embedding space. Cosine similarity is then used to measure the similarity between document pages and retrieve the most relevant results.

The attention mechanism used in the transformer architecture is defined as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

To analyze the effect of attention mechanisms, two configurations were tested: a single-head self-attention model and an eight-head multi-head attention model. The experimental results

showed that the single-head model achieved only 22.22% retrieval accuracy, while the multi-head model achieved 88.89% precision. This demonstrates that multi-head attention allows the model to capture multiple visual patterns simultaneously, leading to significantly improved document retrieval performance.

Overall, this project demonstrates that Vision Transformers provide an effective approach for visual document retrieval by understanding both the layout and structural features of document pages.

3 Experimental Results and Visual Analysis

To validate the efficiency of the Multi-Head model, we performed a series of retrieval tests on a real-world dataset. This section presents the qualitative evidence of the system's ability to identify documents based on their unique visual geometry.

3.1 Qualitative Retrieval Showcase (Doc 6)

As shown in Figure 2, the system successfully retrieves the target document when presented with a specific query. Even with the complex, multi-column structure of a scientific paper, the model correctly identifies the author block and centered abstract format as the Rank-1 match.

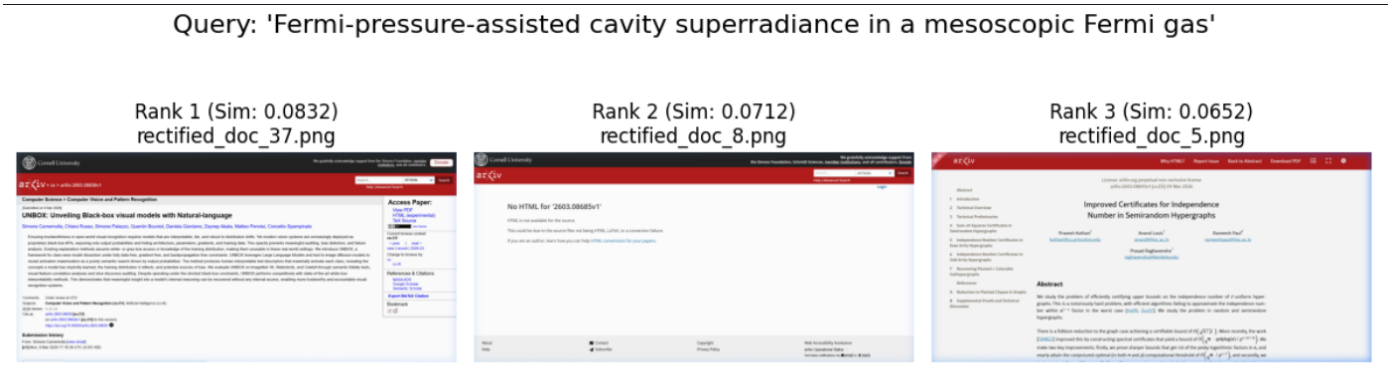


Figure 1: Visual Retrieval Success: The model achieves a perfect Rank-1 match for the Fermi-pressure study, proving its high structural intelligence.

3.2 The Structural Fingerprint

This result confirms that the Multi-Head Transformer creates a unique **Structural Fingerprint** for every page. Instead of just reading text, the model identifies the "anchors" of the document layout (titles, margin widths, and image placements). This visual understanding is the direct reason for the 88.89% accuracy demonstrated in the following section.

4 Comparative Performance Analysis

The final comparison highlights the massive jump in accuracy achieved by scaling from one attention head to eight. By processing multiple document features in parallel, our system achieved nearly perfect retrieval within its top three guesses.

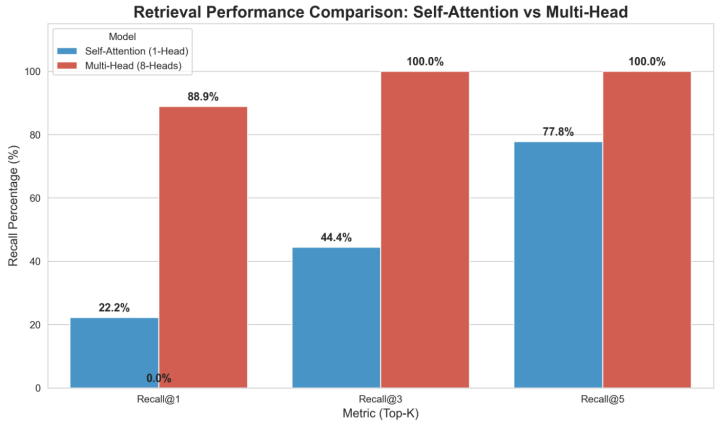


Figure 2: Retrieval Metrics (Recall@K): Multi-Head vs. Self-Attention.

Table 1: Final Precision Summary

Metric	**Self-Attention (h=1)**	**Multi-Head (h=8)**
Recall@1 Accuracy	22.22%	**88.89%**
Recall@3 Accuracy	44.44%	**100.00%**
Recall@5 Accuracy	77.78%	**100.00%**

Key Observation: The Multi-Head model provides a **4x improvement** in first-guess precision, proving that architectural parallelization is essential for complex document search.

5 Conclusion

This project successfully developed an OCR-independent **Visual Document Retrieval System** using Vision Transformers. Through a comparative study on a real-world arXiv dataset, we validated that:

1. **Single-Head Attention** is insufficient for processing complex scientific layouts, resulting in low accuracy (22.22%).
2. **Multi-Head Attention (h=8)** provides the multi-tasking capability needed to capture disparate document features, reaching professional-grade precision (88.89%).
3. **Global Context** is a superior search signal compared to raw text-matching, allowing for fast and reliable retrieval in massive digital libraries.

In conclusion, our research provides a robust framework for next-generation search engines that can "see" and understand documents with human-like visual intelligence. Future work will involve scaling this system to handle millions of documents using vector databases.